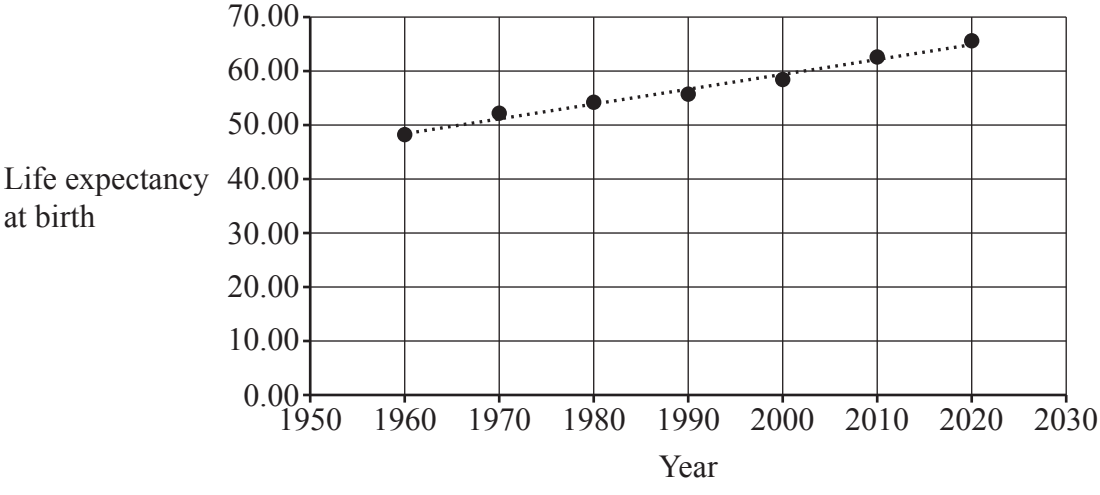
10 The pre-release material contains information about life expectancy at birth for countries of the world at 10-year intervals from 1960 until 2020.

The life expectancy at birth of the population of Sudan for this period, together with a line of best fit, is shown in **Fig. 10.1**.

Fig. 10.1



The equation of the line of best fit is $L = 0.275X - 490$, where X is the year and L is the life expectancy at birth, measured in years.

(a) Use the equation of the line of best fit to estimate the life expectancy at birth in Sudan in 1975. **[1]**

(b) Explain whether this estimate is likely to be close to the true value of life expectancy at birth in Sudan in 1975. **[1]**

(c) Use your knowledge of the pre-release material to explain whether your answer to part **(a)** is likely to be a close approximation to the life expectancy at birth in the United Kingdom in 1975. **[1]**

The pre-release material also gives the median age of the population in countries of the world.

The table shows the median age of the population and the life expectancy at birth in 2020 for some countries in Africa.

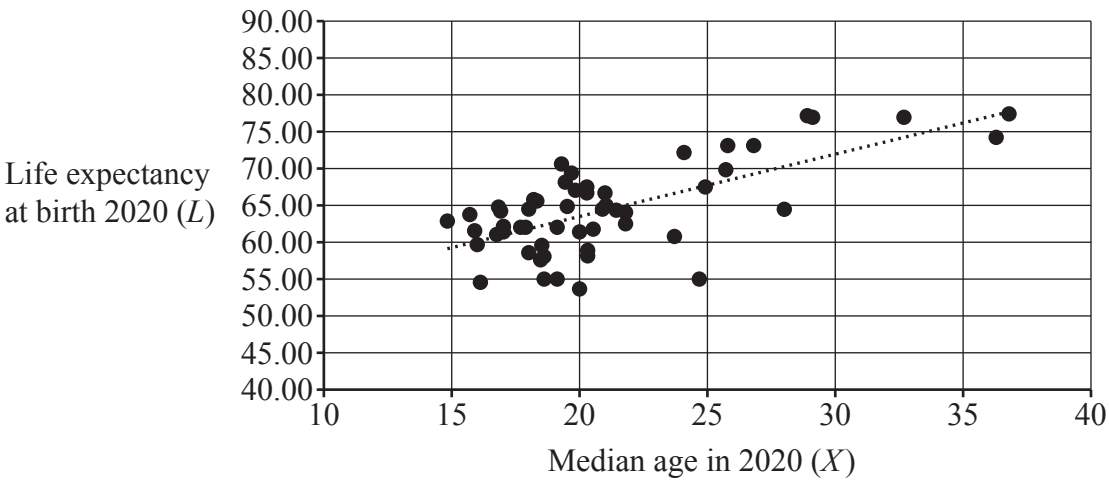
Country	Median age	Life expectancy at birth 2020
Nigeria	18.6	55.02
Rwanda	19.7	69.33
Saint Helena, Ascension and Tristan da Cunha	43.2	#N/A
Sao Tome and Principe	19.3	70.58
Senegal	19.4	68.21

(d) Explain how the data in the table should be cleaned before these data can be included in a scatter diagram for life expectancy at birth against median age for the countries in Africa. **[1]**

Fig. 10.2 shows a scatter diagram for the cleaned data of life expectancy at birth against median age in 2020 for the countries in Africa. It also shows a line of best fit.

The product moment correlation coefficient for these data is 0.680.

Fig. 10.2



(e) A student decides to use the line of best fit to estimate the life expectancy at birth in 2020 for Saint Helena, Ascension and Tristan da Cunha.

Explain whether this estimate is likely to be reliable. **[1]**

10(a)	
10(b)	
10(c)	
10(d)	
10(e)	